

Problem Set: Causality and Potential Outcomes

BUSS975: Causal Inference in Financial Research

Problems 1–9 require no software (except 8(g)). Problem 10 asks you to verify a calculation in R (or Stata); attach your code.

Problem 1 (Description, Prediction, or Causation?)

For each statement below, (i) classify the underlying question as **descriptive**, **predictive**, or **causal**, and (ii) if the statement draws a causal conclusion, name which of the three fallacies from lecture it commits — prediction mistaken for causation, *post hoc ergo propter hoc*, or “no correlation implies no causation.”

- (a) “Firms with more independent boards have higher Tobin’s Q, so appointing independent directors raises firm value.”
- (b) “What fraction of KOSPI firms repurchased shares in 2025?”
- (c) “Our model forecasts next quarter’s default rate with 91% accuracy, so the variables in it are the drivers of default.”
- (d) “The Bank of Korea cut rates in March and GDP growth rose in Q2. The rate cut worked.”
- (e) “Across firms, currency hedging is uncorrelated with the probability of financial distress. Hedging therefore does not reduce distress risk.”
- (f) For part (e), describe the assignment mechanism that would generate a zero correlation *even if* hedging strongly reduced distress. Which lecture example does this parallel?

Problem 2 (Potential Outcomes and the Decomposition)

A banking regulator can inject capital into distressed banks. Let $D_i = 1$ if bank i receives an injection, and let the outcome Y_i be lending growth (%) over the following year. The complete (and, in reality, unobservable) potential outcomes for ten banks are:

Bank i	1	2	3	4	5	6	7	8	9	10
Y_i^0	1	2	3	2	4	7	8	9	8	10
Y_i^1	8	8	8	6	7	9	9	9	7	8

The regulator has perfect foresight and injects capital into the five banks with the **largest treatment effects**.

- Compute δ_i for each bank. Which five banks are treated?
- Compute the true ATE , ATT , and ATU . Verify that $ATE = \pi \cdot ATT + (1 - \pi) \cdot ATU$, where π is the share treated.
- Write out the observed data table — that is, the Y_i an econometrician actually sees, together with D_i . Which entries of the potential outcomes table are missing?
- Compute the simple difference in outcomes,

$$SDO = E[Y^1 | D = 1] - E[Y^0 | D = 0].$$

- Compute the selection bias term and the heterogeneous treatment effect bias term, and verify numerically that

$$SDO = ATE + \underbrace{\left(E[Y^0 | D=1] - E[Y^0 | D=0] \right)}_{\text{selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{heterogeneous TE bias}}.$$

- The true ATE is positive but the SDO is negative. Explain in words — as you would to a journalist — why a well-intentioned, *competent* regulator produces data suggesting its own program was harmful.

Problem 3 (The Assignment Mechanism Is Everything)

Use the same ten banks and the same potential outcomes as Problem 2. Now suppose a **different** regulator is in charge, one who injects capital into banks 6–10 (the five healthiest banks) — perhaps because they are the most politically connected, or because the regulator wants to back “winners.”

- Recompute ATT , ATU , and SDO under this assignment.
- Recompute the selection bias and heterogeneous treatment effect bias terms, and verify the decomposition again.
- You now have three numbers describing the *same* ten banks with the *same* treatment effects: the true ATE , the SDO under the Problem 2 assignment, and the SDO under the Problem 3 assignment. Present them in a small table and comment.
- Note that ATE did not change between Problems 2 and 3, but ATT did. Explain why ATE is a property of the population while ATT is a property of the population *and* the assignment mechanism.
- A colleague proposes: “The problem in Problem 2 is that the sample is too small. With 10,000 banks the bias would disappear.” Is this correct? Explain.

Problem 4 (Randomization and Independence)

Suppose that instead of the regulators in Problems 2 and 3, treatment is assigned by lottery, so that

$$(Y^1, Y^0) \perp\!\!\!\perp D.$$

- (a) State precisely what this independence condition means in words. Why is it stated in terms of *potential* outcomes rather than observed outcomes?
- (b) Using independence, show that $E[Y^0 \mid D = 1] = E[Y^0 \mid D = 0]$ and hence that the selection bias term is zero.
- (c) Show that independence also implies $ATT = ATU = ATE$, so that the heterogeneous treatment effect bias term is zero.
- (d) Conclude that $SDO = ATE$ under randomization. Which of the two bias terms would still vanish if treatment were independent of Y^0 but *not* of δ_i ? Give a finance example in which treatment is plausibly independent of baseline outcomes but not of gains.
- (e) Randomization balances covariates *in expectation*. In any one realized experiment, treated and control groups will still differ somewhat on observables. Does this imply the estimate is biased? Explain the distinction between bias and sampling variability.
- (f) A referee writes: “The authors’ randomized experiment reports a balance table in which 2 of 30 covariates differ at the 5% level. This casts doubt on the randomization.” Evaluate this criticism.

Problem 5 (SUTVA in Financial Markets)

For each scenario, state whether the violation is of **no interference** or **no hidden variation in treatment**, explain the mechanism, and state whether the naive treated-minus-control comparison would likely **overstate** or **understate** the true effect of treatment on the treated.

- (a) A government recapitalizes half the banks in a concentrated banking market. Treated banks use the capital to bid aggressively for deposits and creditworthy borrowers, drawing both away from untreated banks.
- (b) A researcher studies “the effect of receiving TARP funds” on bank lending. TARP investments in the sample range from \$200 million to \$25 billion, and larger banks received larger injections.
- (c) A subsidy is given to a random 30% of firms in an industry. The industry’s total output is fixed by demand, so subsidized firms gain market share exactly at the expense of unsubsidized firms.
- (d) A study estimates the effect of “adopting an anti-takeover provision” on firm value, pooling classified boards, poison pills, and supermajority requirements into a single indicator.
- (e) In (c), the researcher’s DiD finds a large positive effect on treated firms’ output. Is the estimate wrong? Distinguish carefully between the *partial-equilibrium* and *general-equilibrium* effects, and state which question each answers.
- (f) Explain why SUTVA is logically prior to the independence assumption — that is, why a SUTVA violation is a more fundamental problem than selection bias.

Problem 6 (The Regression Bridge: Omitted Variables)

- (a) You estimate $y = \beta_0 + \beta_1 x + u$ by OLS when the true model is $y = \beta_0 + \beta_1 x + \beta_2 z + v$ with $E[v \mid x, z] = 0$. Show that

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \delta_{xz} \beta_2, \quad \delta_{xz} = \frac{\text{cov}(x, z)}{\text{var}(x)}.$$

- (b) For each case below, sign the bias in $\hat{\beta}_1$ and justify the two sign assumptions you need:
- y = firm value, x = governance index, omitted z = CEO talent.
 - y = firm value, x = leverage, omitted z = investment opportunities.
 - y = loan default, x = relationship lending, omitted z = borrower quality observed by the bank but not the econometrician.
- (c) Now connect the two dialects. Consider regressing the observed outcome on a treatment dummy:

$$Y_i = \alpha + \delta D_i + u_i.$$

Using the switching equation $Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$ and writing $\delta = E[\delta_i]$, show that the error term is

$$u_i = (Y_i^0 - E[Y^0]) + D_i (\delta_i - \delta),$$

and hence that $\text{cov}(D, u) \neq 0$ precisely when there is selection bias or selection on gains.

- (d) In one or two sentences: why is it accurate to say that in causal settings, omitted variable bias *is* selection bias rather than merely resembling it?

Problem 7 (Two Channels of Selection)

The error term from Problem 6(c) has two pieces, and D can be correlated with either:

$$u_i = \underbrace{(Y_i^0 - E[Y^0])}_{\text{baseline}} + \underbrace{D_i (\delta_i - \delta)}_{\text{gain}}$$

- (a) Define **selection on levels** and **selection on gains** in terms of which piece D is correlated with. State what each one means substantively — what is the treatment being assigned on?
- (b) Classify each of the following. Some are both.
- A regulator recapitalizes the weakest banks in a crisis.
 - A regulator recapitalizes the banks where an injection will do the most good.
 - Firms adopt a governance provision when their managers expect the largest improvement in value.
 - Firms with a history of poor performance are targeted by activist investors.
- (c) Return to the Perfect Regulator of Problem 2, which treated the five banks with the **largest** δ_i . Which channel is that? Explain why defining selection bias as “ $\text{cov}(D, Y^0) \neq 0$ ” would misclassify it.
- (d) Construct a numerical example — four units is enough — in which $\text{cov}(D, Y^0) = 0$ exactly, yet the naive comparison is still biased for the ATE . What is the source of the bias?

Problem 8 (A Lemma for Binary Treatment)

- (a) **Prove the lemma.** For binary D with $\pi = P(D=1)$ and any random variable W ,

$$\text{cov}(D, W) = \pi(1 - \pi)(E[W | D=1] - E[W | D=0]).$$

Hint: $E[DW] = \pi E[W | D=1]$ because D is binary; then use the law of total expectation for $E[W]$.

- (b) Apply the lemma with $W = Y^0$ to obtain $\text{cov}(D, Y^0)$ in terms of conditional means.
- (c) Apply it with $W = D_i(\delta_i - \delta)$. Show first that $E[W | D=0] = 0$ and $E[W | D=1] = ATT - ATE$, then write down $\text{cov}(D, W)$.
- (d) Note that $\text{var}(D) = \pi(1 - \pi)$ for binary D . Divide each covariance by $\text{var}(D)$ and say what each quotient equals.
- (e) **Put it together.** Using $\hat{\beta}_1 \xrightarrow{p} \delta + \text{cov}(D, u)/\text{var}(D)$ and the identity $ATT - ATE = (1 - \pi)(ATT - ATU)$ from Problem 2, show that

$$\hat{\beta}_1 \xrightarrow{p} ATE + \underbrace{(E[Y^0|D=1] - E[Y^0|D=0])}_{\text{selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{heterogeneity bias}}$$

- (f) Compare the right-hand side with your answer to Problem 2(e). What have you shown about the relationship between the regression route and the potential-outcomes route?
- (g) **Verify numerically.** Using the ten banks and the treatment assignment of Problem 2, compute $\text{cov}(D, Y^0)$, $\text{cov}(D, D_i(\delta_i - \delta))$, and $\text{var}(D)$ directly. Confirm that dividing by $\text{var}(D)$ reproduces the two bias terms, and that the three terms sum to the *SDO*. Then run `lm(Y ~ D)` on the observed data and check that the coefficient matches.
- (h) A referee writes: “Your OLS estimate is inconsistent, so the paper tells us nothing.” Give the precise sense in which $\hat{\beta}_1$ is consistent, and explain why that does not rescue the paper.

Problem 9 (Simultaneity, Bad Controls, and Heterogeneous Effects)

- (a) Consider the simultaneous system

$$y = \beta_0 + \beta_1 x + u, \quad x = \alpha_0 + \alpha_1 y + \varepsilon,$$

with u and ε mutually uncorrelated and $\alpha_1 \beta_1 \neq 1$. Solve for the reduced form of x and show that $\text{cov}(x, u) \neq 0$ whenever $\alpha_1 \neq 0$. What is the sign of the bias in $\hat{\beta}_1$ if $\alpha_1 > 0$?

- (b) A researcher, aware of reverse causality, replaces x_t with x_{t-1} . Explain why this does not generally solve the problem. Under what condition *would* lagging be sufficient?
- (c) **Bad controls.** Suppose x_1 is randomly assigned but x_2 is affected by x_1 :

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \quad x_2 = \gamma_0 + \gamma_1 x_1 + \varepsilon.$$

Explain what $\hat{\beta}_1$ estimates when x_2 is included, and why this is not the causal effect of x_1 on y even though x_1 was randomized. Give a corporate finance example.

- (d) **Random coefficients.** Let each firm have its own intercept and slope:

$$y_i = a_i + b_i x_i, \quad a_i = \alpha + c_i, \quad b_i = \beta + d_i, \quad E[c_i] = E[d_i] = 0.$$

Rewrite this as a regression of y_i on x_i and identify the composite error term. State the two conditions required for OLS to consistently estimate the average partial effect $\beta = E[b_i]$.

- (e) With binary x_i , map the random coefficients model onto the potential outcomes framework: what are a_i and b_i in potential outcomes notation, and what do the two conditions in (d) correspond to in the language of Problem 2?
- (f) A paper reports an average effect and separately argues at length that the effect is much stronger for financially constrained firms. Explain the tension between these two claims, and state how the paper should phrase its interpretation of the average coefficient.

Problem 10 (Randomization Inference)

Six firms are enrolled in a pilot program. Three are selected **by lot** to receive treatment. The realized assignment and observed outcomes are:

Firm i	1	2	3	4	5	6
D_i	1	1	1	0	0	0
Y_i	6	7	8	3	4	5

- (a) Compute the observed SDO .
- (b) State Fisher's sharp null hypothesis. Explain how it differs from the null that the *average* treatment effect is zero, and why the sharp null — but not the average null — allows you to fill in every missing potential outcome.
- (c) How many distinct assignments of three treated firms out of six are possible? Under the sharp null, explain why the observed outcomes Y_i would have been identical under every one of them.
- (d) Without enumerating all of them, identify the assignment that produces the **largest** possible SDO and the one that produces the smallest (most negative). What does this tell you about where the realized assignment sits in the permutation distribution?
- (e) Compute the exact two-sided p -value, defined as the share of assignments for which $|SDO| \geq |SDO_{\text{observed}}|$.
- (f) The realized assignment was the most extreme of all possible assignments, yet the p -value does not fall below 0.05. Explain what this implies about the statistical power of very small experiments, and what the smallest attainable two-sided p -value would be with six units and three treated.
- (g) **Computational.** Verify your answer to (e) in R (or Stata). A skeleton is given below; report your code and output.

```

y <- c(6, 7, 8, 3, 4, 5)
treated <- c(1, 2, 3)
sdo_obs <- mean(y[treated]) - mean(y[-treated])

perms <- combn(6, 3)          # all possible assignments (columns)
sdo_all <- apply(perms, 2, function(tr) mean(y[tr]) - mean(y[-tr]))

# exact two-sided p-value
mean(abs(sdo_all) >= abs(sdo_obs))

```

- (h) Explain why randomization inference is particularly valuable in finance settings with one treated state, one regulatory event, or a handful of treated firms — and name one method later in the course that relies on exactly this logic.